

Pulmonary Hypertension — Diagnostic Workup & Vasoreactivity



1. Only IPAH/HPAH tested

WHAT YOU SEE

Banner: EXCLUSIVE - ONLY IPAH / HEREDITARY PAH, velvet rope

WHAT IT ENCODES

Acute VASOREACTIVITY testing is done ONLY in idiopathic (IPAH), heritable (HPAH), or drug/toxin-induced PAH — the subsets where a small minority are 'responders' who benefit from calcium channel blockers. It is performed at the time of the diagnostic right heart catheterization, not by echo. It is NOT used to guide therapy in connective-tissue-disease PAH or other PAH subtypes.

BOARD TRAP

WRONG: performing vasoreactivity testing in Group 2, 3, or 4 PH. Discriminator: vasoreactivity testing and CCB therapy apply only to IPAH/HPAH/drug-induced Group 1 — in Group 2 (left heart) a positive 'response' is meaningless and CCBs can be harmful.

2. Groups 2/3/5 barred

WHAT YOU SEE

Group 2, 3, 5 figures behind rope with red X

WHAT IT ENCODES

Groups 2 (left heart), 3 (lung disease), and 5 (multifactorial) are NOT vasoreactivity-tested and do not receive calcium channel blockers on that basis. Their PH is driven by the underlying disorder — treat heart failure/valve, hypoxia/lung disease, or the systemic condition respectively.

BOARD TRAP

WRONG: extrapolating a 'positive vasoreactivity' result to start CCBs in a Group 2 or 3 patient. Discriminator: CCB responsiveness is only validated and only safe in IPAH/HPAH/drug-induced PAH; in left-heart or lung disease, treat the cause.

3. CCB reward 5%

WHAT YOU SEE

Trophy labeled CCB REWARD FOR 5%

WHAT IT ENCODES

Only about 10% of IPAH patients are acute responders, and just ~5% maintain a durable long-term response to high-dose CALCIUM CHANNEL BLOCKERS (nifedipine, diltiazem, or amlodipine). Responders must be re-evaluated within 3-12 months to confirm sustained benefit; non-sustained responders need PAH-specific therapy.

BOARD TRAP

WRONG: starting calcium channel blockers empirically in all PAH patients. Discriminator: only the ~5% durable vasoresponders benefit; CCBs in non-responders cause hypotension, edema, and clinical deterioration — confirm a positive test AND sustained response first.

4. mPAP drop ≥ 10

WHAT YOU SEE

Scoreboard: mPAP DROP ≥ 10 mmHg green check

WHAT IT ENCODES

A POSITIVE acute vasoreactivity response (modified Sitbon criteria) requires a FALL in mPAP of AT LEAST 10 mmHg during the challenge. This component must be met together with the absolute-mPAP and preserved-cardiac-output criteria.

BOARD TRAP

WRONG: calling any drop in mPAP a positive test. Discriminator: a positive result needs a fall of ≥ 10 mmHg AND a final mPAP ≤ 40 mmHg AND maintained/increased cardiac output — all three, not just a partial pressure drop.

5. Absolute mPAP ≤ 40

WHAT YOU SEE

Scoreboard: ABSOLUTE mPAP ≤ 40 mmHg green check

WHAT IT ENCODES

A positive response also requires the FINAL absolute mPAP to fall TO ≤ 40 mmHg. A patient can drop ≥ 10 mmHg yet still be non-responsive if the ending mPAP remains above 40.

BOARD TRAP

WRONG: declaring a responder based on the ≥ 10 mmHg drop alone when the final mPAP is still > 40 . Discriminator: both the magnitude of fall (≥ 10) AND the absolute endpoint (≤ 40 mmHg) must be satisfied for a positive test.

6. CO must not drop

WHAT YOU SEE

Lightning bolt text CO MUST NOT DROP

WHAT IT ENCODES

During the vasoreactivity test, CARDIAC OUTPUT must be MAINTAINED or INCREASED. A drop in CO means the agent caused systemic vasodilation/hemodynamic compromise rather than selective pulmonary vasodilation — that is a NEGATIVE (and concerning) response.

BOARD TRAP

WRONG: calling the test positive when mPAP falls but cardiac output also drops. Discriminator: a falling CO indicates harmful systemic effect, not true pulmonary vasoreactivity — the result is negative and CCBs would be dangerous.

7. RHC = gold standard

WHAT YOU SEE

Patient on track with trophy labeled GOLD STANDARD - RHC

WHAT IT ENCODES

RIGHT HEART CATHETERIZATION is the GOLD-STANDARD confirmatory test for PH: it directly measures mPAP, PAWP, cardiac output (thermodilution/Fick), and calculates PVR — the numbers that both confirm PH and classify pre- vs post-capillary disease. It is required before committing a patient to PAH-specific therapy.

BOARD TRAP

WRONG: confirming PH (or starting PAH drugs) on the basis of echocardiography alone. Discriminator: echo only screens/estimates PA systolic pressure; right heart catheterization is mandatory to confirm the diagnosis, get the PAWP, and classify the hemodynamics.

8. Adenosine/NO agents

WHAT YOU SEE

Inhaled NO canister and adenosine bottle, INHALED iloprost stand

WHAT IT ENCODES

Vasoreactivity is tested with SHORT-ACTING, rapidly reversible pulmonary vasodilators: inhaled NITRIC OXIDE (10-20 ppm, the most commonly used), IV ADENOSINE, or IV epoprostenol (and in some labs inhaled iloprost). Their short half-lives make the response immediately interpretable and reversible if hemodynamics worsen.

BOARD TRAP

WRONG: using oral calcium channel blockers themselves as the acute testing agent. Discriminator: testing uses titratable short-acting agents (inhaled NO, IV adenosine, IV epoprostenol); CCBs are the long-term therapy given only AFTER a confirmed positive test.

9. Echo + bubble study

WHAT YOU SEE

Echo screen labeled ECHO + BUBBLE STUDY

WHAT IT ENCODES

Transthoracic ECHOCARDIOGRAPHY is the FIRST-LINE SCREENING test for suspected PH: it estimates PA systolic pressure from the tricuspid regurgitant jet, assesses RV size/function and the interventricular septum, and evaluates left-heart causes (Group 2). A bubble (agitated-saline) study detects an intracardiac/right-to-left SHUNT (e.g., PFO/ASD).

BOARD TRAP

WRONG: treating echo as diagnostic/confirmatory for PH. Discriminator: echo SCREENS and stratifies probability — an abnormal echo triggers right heart catheterization, which is what actually confirms and classifies PH.

10. Isolated low DLCO

WHAT YOU SEE

PFT screen: ISOLATED low DLCO = PAH clue

WHAT IT ENCODES

On pulmonary function testing, an ISOLATED reduction in DLCO (diffusing capacity) with relatively PRESERVED spirometry/lung volumes is a classic clue to pulmonary VASCULAR disease (PAH) — the capillary bed is destroyed/remodeled even though the airways and parenchyma are intact. A very low DLCO in PAH also flags possible pulmonary veno-occlusive disease and portends worse prognosis.

BOARD TRAP

WRONG: dismissing PAH because spirometry (FEV1/FVC) is normal. Discriminator: PAH characteristically gives an ISOLATED low DLCO with normal volumes — obstructive/restrictive PFT abnormalities instead point toward Group 3 lung-disease PH.

11. V/Q scan for CTEPH

WHAT YOU SEE

V/Q SCAN screen labeled SCREEN CTEPH

WHAT IT ENCODES

The VENTILATION/PERFUSION (V/Q) scan is the SCREENING test of choice for chronic thromboembolic PH (CTEPH, Group 4): it shows segmental/subsegmental PERFUSION defects with normal ventilation (mismatched defects). A normal V/Q scan essentially excludes CTEPH. Every PH patient should be screened for CTEPH because it is potentially CURABLE by pulmonary thromboendarterectomy.

BOARD TRAP

WRONG: using CT pulmonary angiography to rule OUT CTEPH. Discriminator: CTPA can miss distal/chronic organized clot — V/Q scan is more SENSITIVE and is the screening test; a normal V/Q reliably excludes CTEPH, whereas a normal CTPA does not.

12. TR murmur / TR jet

WHAT YOU SEE

Auscultation figure: holosystolic TR murmur, increases with inspiration

WHAT IT ENCODES

A holosystolic tricuspid REGURGITATION murmur at the left lower sternal border that gets LOUDER with INSPIRATION (Carvallo sign, due to increased venous return) reflects RV pressure/volume overload from PH. Other exam clues: a loud/palpable P2, RV heave, elevated JVP with prominent V waves, and signs of right heart failure (edema, ascites, hepatomegaly).

BOARD TRAP

WRONG: assuming an inspiration-augmented holosystolic murmur is mitral regurgitation. Discriminator: right-sided murmurs (TR) increase with INSPIRATION (Carvallo sign); left-sided murmurs (MR) do not — the inspiratory augmentation localizes it to the tricuspid/right heart.

13. Elevated TR jet velocity

WHAT YOU SEE

Echo tag ELEVATED TR JET, enlarged RA/RV

WHAT IT ENCODES

On echo, an elevated tricuspid regurgitant JET VELOCITY is used to estimate PA systolic pressure (via the modified Bernoulli equation, adding estimated RA pressure). A high TR velocity, plus enlarged RA/RV and septal flattening, raises the echo PROBABILITY of PH and triggers confirmatory right heart catheterization.

BOARD TRAP

WRONG: quoting the echo-estimated PA systolic pressure as the patient's actual mPAP/diagnosis. Discriminator: TR-jet-derived PASP is an estimate with significant error — it stratifies probability and prompts catheterization; only RHC gives the true mPAP and PAWP.

14. High DLCO = classic PH

WHAT YOU SEE

Bottom tag reading DLCO down-arrow = CLASSIC PH (the diffusing capacity arrow points DOWN) — the falling DLCO IS the fingerprint of pulmonary vascular disease.

WHAT IT ENCODES

Pattern recognition for boards: an ISOLATED LOW DLCO with PRESERVED lung volumes is the hallmark PFT signature of pulmonary VASCULAR disease/PAH. (Contrast: a HIGH DLCO classically accompanies alveolar hemorrhage, polycythemia, left-to-right shunt, or asthma.) Pair the low DLCO with normal spirometry to point toward Group 1 PAH over parenchymal lung disease.

BOARD TRAP

WRONG: expecting PAH to show a HIGH DLCO. Discriminator: PAH gives a LOW (often disproportionately low) DLCO with normal volumes; a HIGH DLCO instead suggests alveolar hemorrhage, polycythemia, or left-to-right shunt — different differential entirely.